

Data Explorer

Overview

In this project, you'll explore Python concepts like loops, lists, functions, and file handling to analyze data. You'll get a sense of what data scientists and analysts do in their daily work.

Resources

- You can download optional files to help you. Choose a genre and download the dataset from the link:
<https://www.kaggle.com/datasets/rajugc/imdb-movies-dataset-based-on-genre>

Project Instructions

Part 1: Building Foundations with Lists and Loops

Session 1: Introduction to Lists and Loops

- **Objective:** Learn how to create and manage lists and use loops to work with data.

Activities:

1. **Introduction to Lists:**
 - **Concepts:** Understand what lists are, how to create them, and perform basic operations like adding, accessing, and modifying elements.
 - **Task:** Create a list of movie titles, ratings, and genres. Practice accessing and updating elements in the list.
2. **Introduction to Loops:**
 - **Concepts:** Learn the basics of `for` and `while` loops and how they iterate over lists.
 - **Task:** Write a loop to print each movie title and calculate the sum of ratings.
3. **Combined Practice:**
 - **Exercise 1:** Write a loop that prints both the movie title and its rating from the list.
 - **Exercise 2:** Find the highest-rated movie using a loop.
 - **Exercise 3:** Create a loop that filters movies by genre and stores them in a new list.
 - **Discussion:** Reflect on how loops and lists are essential for handling and processing data.

Session 2: Introduction to Functions

- **Objective:** Learn how to define and use functions to organize and reuse code.

Activities:

1. **Introduction to Functions:**
 - **Concepts:** Understand the purpose of functions, how to define them, use parameters, and return values.
 - **Examples:** Create simple functions, such as one that calculates the sum of two numbers or prints a greeting.
 2. **Practice with Functions:**
 - **Exercise 1:** Write `calculate_average_rating(ratings)` to compute the average rating from a list.
 - **Exercise 2:** Write `find_highest_rated(movies)` to find the highest-rated movie using a loop.
 3. **Application and Integration:**
 - **Activity:** Combine functions with lists and loops:
 - Use `calculate_average_rating()` on a list of ratings.
 - Apply `find_highest_rated()` to a list of movies to find the top-rated one.
 - Implement `filter_by_genre()` to filter and analyze the movie dataset.
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Part 2: Expanding Skills and Analysis

Session 3: File Handling

- **Objective:** Learn how to read from and write to files for data persistence and analysis.

Activities:

1. **Introduction to File Handling:**
 - **Concepts:** Learn the basics of opening, reading, writing, and closing files.
 - **Examples:** Show how to read data from a text file and write data to a text file.
2. **Practice with File Handling:**
 - **Task 1:** Download a dataset from the [link](#).
 - **Task 2:** Write `load_movies(filename)` to read movie data from a file into a list.
 - **Task 3:** Write `save_analysis(results, filename)` to save analysis results to a text file.



Mission Bit

- **Task 4:** Practice reading a movie dataset from a file and processing it using the functions created earlier.
- 3. **Application:**
 - **Activity:** Load a movie dataset from a file, analyze it using your functions (like calculating the average rating and finding the highest-rated movie), and save the results to a new file.
 - **Note:** You may need to update your functions to match the format of the new dataset. Make sure they can handle different types of data.

Session 4: Open Work Time

- **Objective:** Use this time to work on your project, catch up, or explore further.

Activities:

1. **Independent Work:**
 - Continue working on your movie data analysis project.
 - Apply any feedback received earlier to improve your functions, file handling, and data analysis.
 - Explore additional challenges, such as creating more complex summaries or working with different datasets.
2. **Student Presentations:**
 - Share your project progress and findings with the class.

Project Workflow Summary:

- **End of Week 1:**
 - You'll have a solid understanding of lists, loops, and functions.
 - You'll be able to process data with loops and lists and perform basic calculations using functions.
- **End of Week 2:**
 - You'll be skilled in file handling and advanced data analysis.
 - You'll be able to analyze datasets from files, generate insights, and save your results.

Extensions:

 Project 2 Extra: Conditionals at your own risk.

